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Graded Lab: Section F

Problem Statement: CNN-Based Image Classification (5 Marks)

Download a subset of the CIFAR-10 dataset (or use full dataset if resources allow). Build a Convolutional Neural Network (CNN) to classify images into 10 categories.

1. Data Preparation

- Dataset: CIFAR-10
- Input shape: 32 × 32 × 3 (RGB images)
- Normalize pixel values (0–1 range)
- Split into training and testing sets

2. Build CNN Model

Design a CNN with the following structure:

- Conv Layer 1 → ReLU → MaxPooling
- Conv Layer 2 → ReLU → MaxPooling
- Flatten layer
- Dense Layer (Fully Connected)
- Output Layer (Softmax for 10 classes)

3. Training

- Loss function: Categorical Crossentropy
- Optimizer: Adam
- Metrics: Accuracy
- Train for 5–10 epochs

4. Evaluation

- Evaluate model on test data
- Exhibit:
    - Test accuracy
    - Loss

5. Visualization

- Plot:

- Training vs validation accuracy
- Training vs validation loss

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Source Code:

```python
import tensorflow as tf
from tensorflow.keras import datasets, layers, models
import matplotlib.pyplot as plt

# ==============================
# 1. DATA PREPARATION
# ==============================

# Load CIFAR-10 dataset
(X_train, y_train), (X_test, y_test) = datasets.cifar10.load_data()

# Normalize pixel values (0-255 → 0-1)
X_train = X_train / 255.0
X_test = X_test / 255.0

# Convert labels to one-hot encoding
y_train = tf.keras.utils.to_categorical(y_train, 10)
y_test = tf.keras.utils.to_categorical(y_test, 10)

print("Training shape:", X_train.shape)
print("Testing shape:", X_test.shape)

# ==============================
# 2. BUILD CNN MODEL
# ==============================

model = models.Sequential()

# Conv Layer 1 → ReLU → MaxPooling
model.add(layers.Conv2D(32, (3,3), activation='relu', input_shape=(32,32,3)))
model.add(layers.MaxPooling2D((2,2)))

# Conv Layer 2 → ReLU → MaxPooling
model.add(layers.Conv2D(64, (3,3), activation='relu'))
model.add(layers.MaxPooling2D((2,2)))
```

```python
# Flatten layer
model.add(layers.Flatten())

# Dense Layer
model.add(layers.Dense(64, activation='relu'))

# Output Layer (Softmax)
model.add(layers.Dense(10, activation='softmax'))

# Show model summary
model.summary()

# ==============================
# 3. TRAINING
# ==============================

model.compile(optimizer='adam',
              loss='categorical_crossentropy',
              metrics=['accuracy'])

history = model.fit(
    X_train, y_train,
    epochs=10,
    validation_data=(X_test, y_test)
)

# ==============================
# 4. EVALUATION
# ==============================

test_loss, test_acc = model.evaluate(X_test, y_test)

print("\nTest Accuracy:", test_acc)
print("Test Loss:", test_loss)

# ==============================
```

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script.py > ...

```python
# ==============================
# 5. VISUALIZATION
# ==============================

# Accuracy plot
plt.figure()
plt.plot(history.history['accuracy'], label='Training Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend()
plt.title("Training vs Validation Accuracy")
plt.show()

# Loss plot
plt.figure()
plt.plot(history.history['loss'], label='Training Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()
plt.title("Training vs Validation Loss")
plt.show()
```

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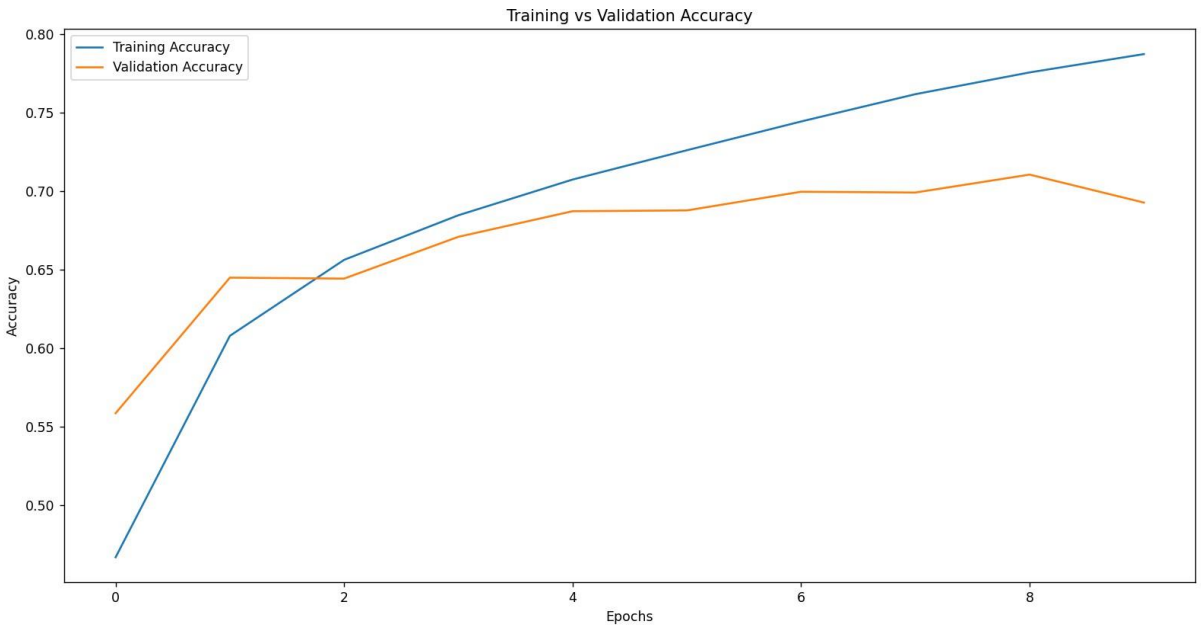
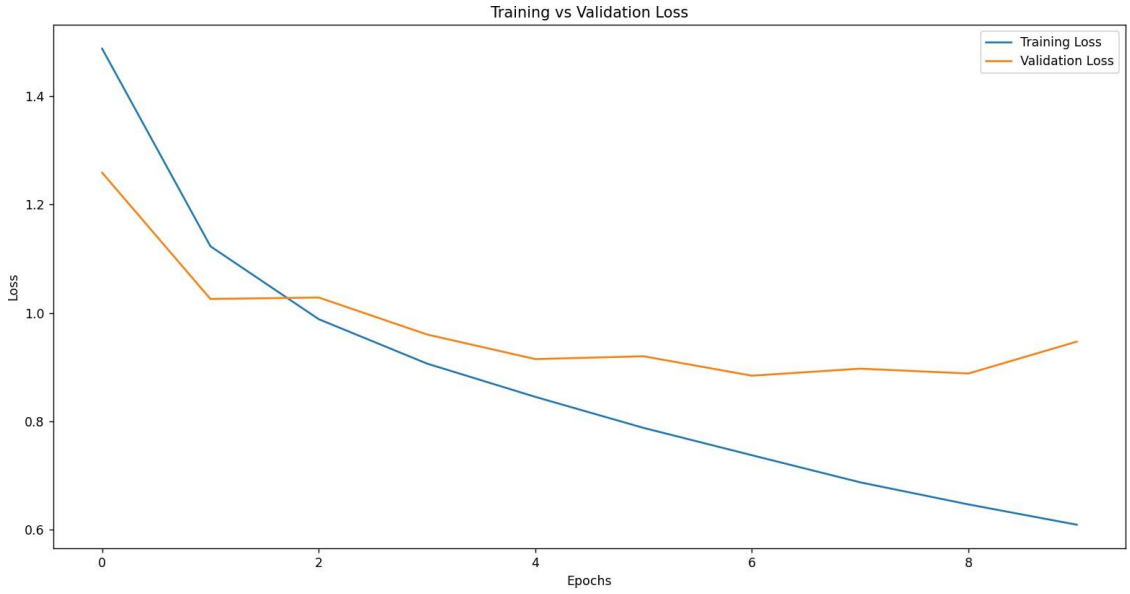
Output:

```
PROBLEMS 3   OUTPUT   DEBUG CONSOLE   ···   Filter (e.g. text, !excludeText, t...   Code

226/313 [====================>.........] - ETA: 0s - loss: 0.9465 - accuracy: 0.6922
234/313 [=====================>........] - ETA: 0s - loss: 0.9426 - accuracy: 0.6940
242/313 [======================>.......] - ETA: 0s - loss: 0.9412 - accuracy: 0.6938
250/313 [======================>.......] - ETA: 0s - loss: 0.9429 - accuracy: 0.6944
258/313 [=======================>......] - ETA: 0s - loss: 0.9421 - accuracy: 0.6948
267/313 [========================>.....] - ETA: 0s - loss: 0.9419 - accuracy: 0.6950
276/313 [=========================>....] - ETA: 0s - loss: 0.9448 - accuracy: 0.6934
285/313 [==========================>...] - ETA: 0s - loss: 0.9438 - accuracy: 0.6944
294/313 [===========================>..] - ETA: 0s - loss: 0.9423 - accuracy: 0.6943
303/313 [============================>.] - ETA: 0s - loss: 0.9436 - accuracy: 0.6944
312/313 [============================>.] - ETA: 0s - loss: 0.9464 - accuracy: 0.6927
313/313 [==============================] - 2s 6ms/step - loss: 0.9469 - accuracy: 0.6926

Test Accuracy: 0.6926000118255615
Test Loss: 0.9469151496887207

[Done] exited with code=0 in 327.599 seconds
```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